

Literature Review:

The relationship between urban design and public transportation accessibility has emerged as a critical issue in recent discussions regarding sustainable urban development. Cities globally face mounting pressure to reconcile urban growth with a wide range of social and environmental issues including impacts on public health, social inequity, road trauma and traffic congestion and environmental pollution. Effective urban design and public transportation planning are powerful and inclusive development tools, capable of providing significant opportunities to mitigate these growing urban problems [1, 2].

Urban design/form—defined as the spatial arrangement of land uses, infrastructure, networks and population distribution—directly impacts transportation demand and accessibility. Historically, transportation planning models prioritized private vehicle use, inadvertently fostering sprawl, congestion, and environmental degradation [1]. However, contemporary urban development plans have begun to emphasize transit-oriented development (TOD), which integrates land use diversity, density, and pedestrian-friendly design to reduce reliance on cars and enhance public transit viability [2]. The interplay between urban design attributes—land use diversity, density, and street design—and transit is bidirectional. While transit infrastructure stimulates denser, mixed-use development around stations (e.g., TOD), well-designed neighborhoods also create demand for transit by concentrating users [1].

1. Land Use Diversity:

Mixed land uses generate overlapping trip purposes (e.g., living, working, shopping), reducing the need for long-distance travel and increasing the feasibility of walking or transit use. Consequently, this creation of relatively self-contained societies reduces the need for internal urban mobility and renders the region easily serviced by public transportation. Studies highlight that neighborhoods with greater housing density, well-connected street networks, and access to a large variety of services see higher transit ridership due to proximity and reduced car dependency [1, 2].

2. Density:

Population, housing and employment density are foundational factors in urban development. High-density residential and business areas concentrate potential riders, enabling frequent service and shorter headways, increasing the use of public transport [1]. Studies have found that communities with residential densities exceeding 20 dwellings per hectare correlates with increased walking, cycling, and transit use, as seen in cities like Melbourne [2]. High-density areas and intensive urban developments broaden the range of services accessible without resorting to automatic travel, reducing journey distances (average distances between homes, services, jobs, etc.) [1].

3. Street Design:

Pedestrian-oriented street networks—characterized by high connectivity, grid-like layouts, and sidewalk infrastructure—enhance the usage and accessibility of public transit. Small, dense street grids create a more accessible environment for pedestrians, reducing distances between land uses and vehicular speeds while fostering safer and more direct routes to transit nodes [1]. Conversely, curvilinear suburban layouts prioritize car travel, fragmenting pedestrian networks [1, 2]. The uses of intersecting streets make walking trips easier, reducing congestion at the entrances to residential areas.

Building on this foundation, this section investigates how variations in urban design—measured through a combination of variables that reflect a location’s land use diversity, density, and/or street design—correlate with public transportation accessibility across NYC’s census block groups. This study aims to assess whether the findings in the articles apply to NYC census block groups and to provide insights for transit-oriented planning in NYC.

Methodology

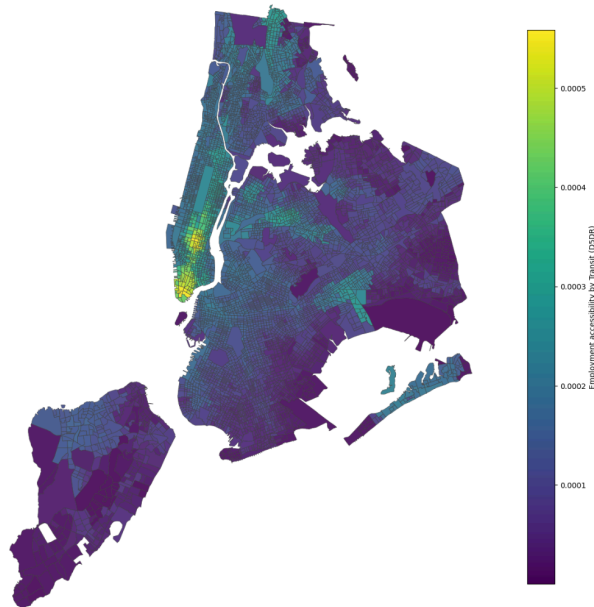
The analysis begins by loading and preprocessing the EPA’s Smart Location Database (SLD) for New York State. The dataset, stored as a geodatabase, is filtered to retain only New York census block groups (STATEFP = '36') and narrowed to five NYC counties (005, 047, 061, 081, 085) to focus on urban areas. Variables are selected to represent urban design (e.g., density, land use diversity, street design) and transit accessibility (D5DR). Metrics like D1A (residential density), D3B (pedestrian-oriented intersections), and D4A (distance to transit) are included to capture physical design elements, while D5DRI (transit centrality) and NatWalkInd (walkability) reflect connectivity. Data cleaning removes invalid rows and enforces constraints to ensure reliability (e.g., negative densities, entropy scores outside 0–1).

A correlation heatmap is generated to quantify relationships between urban design variables and transit accessibility. This step identifies patterns such as the strong positive correlation and anomalies. For instance, the correlation between D5DRI (transit centrality) and D5DR (How easily jobs can be reached via public transit in a specific region compared to the entire Metropolitan Statistical Area) is exactly 1.0. This perfect positive linear correlation is nearly impossible in independent real-world data. This suggests that there likely is some relationship between D5DRI (transit centrality) and D5DR, possibly that D5DR is used in the computation of D5DRI. As such, D5DRI was excluded from the computation of the urban design composite score. While many of the selected variables (NatWalkInd, D2R_JOBPOP, etc.) didn’t have a strong correlation, most having a correlation of approximately 0.0, there still were some variables that had relatively high correlations (D1A, D1B, D1C, etc.) [Reference the Supplement for more information].

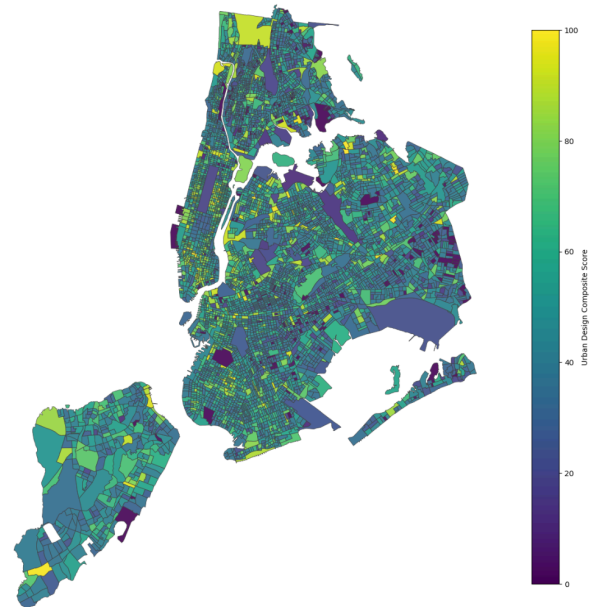
An OLS regression was then run, where a Box-Cox transformation on D5DR to address non-normality and outliers are trimmed using residual quantiles. Model diagnostics (Q-Q plots, VIF scores) were used to ensure residuals are normally distributed and multicollinearity is minimized, which is vital for accurate OLS regression results. Based on the OLS model, statistically significant variables are standardized to create a composite urban design score, scaled to 0–100 for interpretability. This score weights each variable by its regression coefficient, ensuring contributions reflect their impact on transit accessibility. The resulting scores, coupled with D5DR—the Proportional Accessibility of Regional Destinations by Transit (Employment accessibility expressed as a ratio of total MSA accessibility)—and the geometry of each census block group (CBG), are then used to create the following choropleths.

Data Product

Proportional Accessibility of Regional Destinations by Transit
(Employment accessibility by Transit, expressed as a ratio of total MSA accessibility)
in NYC CBGs



Urban Design Composite Score in NYC CBGs



The first choropleth (D5DR: Transit Accessibility) shows the geographic distribution of employment accessibility by transit (D5DR) across census block groups in New York City. Darker colors indicate areas with lower transit accessibility, while lighter colors represent higher accessibility. The second choropleth (Composite Urban Design Score) displays the computed urban design quality composite score (0–100 scale) for the same census block groups, where higher scores (warmer colors) correlate with environments with more urban design attributes that promote employment accessibility by transit. It is interesting to note the wider variety of color in the second choropleth (Composite Urban Design Score), when compared to the first choropleth (D5DR: Transit Accessibility).

Citation

Jasim, Ihsan Abbas, et al. “The Impact of Transit on Sustainable Urban Form.” *IOP Conference Series: Materials Science and Engineering*, vol. 1058, no. 1, 1 Feb. 2021, p. 012049, <https://doi.org/10.1088/1757-899x/1058/1/012049>.

Boulangé, Claire, et al. “Examining associations between urban design attributes and transport mode choice for walking, cycling, public transport and Private Motor Vehicle Trips.” *Journal of Transport & Health*, vol. 6, Sept. 2017, pp. 155–166, <https://doi.org/10.1016/j.jth.2017.07.007>.